After a long sleep, followed by five hours of calculations with general relativity's black-hole formulas, three hours of plowing through Schechter's Black-Hole Atlas, and an hour of consultation with your crew, you formulate the plan for the next leg of your voyage.

Your crew then transmit to the World Geographic Society, under the optimistic assumption that it still exists, an account of your experiences with Sagittario. At the end of their transmission your crew describe your plan:

Your calculations show that the larger the hole, the weaker the rocket blast you will need to support yourself, hovering, at 1.0001 horizon circumferences. For a painful but bearable 10-Earth-gravity blast, the hole must be 15 trillion (15×10^{12}) solar masses. The nearest such hole is the one called Gargantua, far outside the 100,000 (10^5) light-year confines of our own Milky Way galaxy, and far outside the 100 million (10^8) light-year Virgo cluster of galaxies, around which our Milky Way orbits. In fact, it is near the quasar 3C273, 2 billion (2×10^9) light-years from the Milky Way and 10 percent of the distance to the edge of the observable part of the Universe.

The plan, your crew explain in their transmission, is a voyage to Gargantua. Using the usual 1 -*g* acceleration for the first half of the trip and *1-g* deceleration for the second half, the voyage will require 2 billion years as measured on Earth, but, thanks to the speed-induced warpage of time, only 42 years as measured by you and your crew in the starship. If the members of the World Geographic Society are not willing to chance a 4-billion-year deep-freeze hibernation (2 billion years for the starship to reach Gargantua and 2 billion years for its transmission to return to Earth), then they will have to forgo receiving your next transmission.

Gargantua

Forty-two years of starship time later, your ship decelerates into the vicinity of Gargantua. Overhead you see the quasar 3C273, with two brilliant blue jets squirting out of its center[29]; below is the black abyss of Gargantua. Dropping into orbit around Gargantua and making your usual